Mhamed Essafri, Ph.D student in sparse optimization

essafri04@gmail.com Mhamed Essafri
https://messafri.github.io/

Employment History

2023 – 2026 **PhD Researcher, IRIT, Toulouse, France.** Sparse optimization methods for machine learning and signal/image processing; development of exact continuous relaxations for ℓ_0-regularized problems.

2023 **6-month Internship, IRIT, Toulouse, France.** Worked on sparse optimization for machine learning and signal/image processing applications.

Education

2023 – 2026 **Ph.D., INP Toulouse** Informatics and Telecommunication.
Thesis title: *Exact Relaxations for sparse and low-rank optimization*

2021 – 2023 **M.Sc. Applied mathematics, FST Limoges**.
Thesis title: *Mathematical optimization applied to data science and artificial intelligence*.

2018 – 2021 **License. mathematics and applications, FST Settat**.
Thesis title: *Science and technology*.

Research Publications

Journal articles

1. M. Essafri, L. Calatroni, and E. Soubies, "Exact continuous relaxations of ℓ_0-regularized criteria with non-quadratic data terms," *Journal of Global Optimization*, 2025.

Conference Proceedings

1. M. Essafri, L. Calatroni, and E. Soubies, "Box-constrained lo bregman relaxations," in *European Signal Processing Conference (EUSIPCO)*, European Association for Signal Processing (EURASIP), Palermo, Italy, 2025.
2. M. Essafri, L. Calatroni, and E. Soubies, "Regularization path via lo bregman relaxations," in *GRETSI 2025*, Strasbourg, France, 2025.
3. M. Essafri, L. Calatroni, and E. Soubies, "On ℓ_0-bregman-relaxations for kullback-leibler sparse regression," in *IEEE International Workshop on Machine Learning for Signal Processing*, 2024, pp. 1–6.

Talks & Presentations

November 18, 2026 Talk, PGMO Days 2026, Paris, France.

March 18, 2026 Talk, SMAI-MODE 2026, Nice, France.

September 29, 2025 Poster, (Blind) Inverse Problems in Imaging Workshop, Sophia-Antipolis, France.

Talks & Presentations (continued)

September 08, 2025	Poster, 33rd European Signal Processing Conference (EUSIPCO), Palermo, Italy.
August 25, 2025	Talk, GRETSI 2025, Strasbourg, France.
September 22, 2024	Poster, IEEE MLSP 2024, London, UK.
September 16, 2024	Poster, Workshop on ℓ_0-Based Minimization, IHP Workshop, Paris, France.
March 27, 2024	Poster, SMAI-MODE 2024, Lyon, France.
November 28, 2023	Talk, PGMO Days 2023, Paris, France.
November 06, 2023	Talk, ANR Micro-Blind Workshop 2023, CIRM, Marseille, France.

Skills

Languages	Arabic (native), English (fluent), French (fluent)
Coding	Python, C/C++, Matlab, LaTeX
ML & AI	Machine Learning, Deep Learning, Reinforcement Learning, Sparse Optimization, Image and Signal Processing
Tools & Libraries	TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Git
Misc.	Academic research, scientific writing, teaching assistance, numerical optimization, LaTeX typesetting and publishing.

References

Available on Request